

Skills Summary

Finding a Fraction of a Number or Set

Use this reference while completing your worksheet or when studying.

Skill 1 — A unit fraction of a number

Rule: $\frac{1}{d}$ of a number means: split the number into d equal groups and take one group. In symbols, divide by the denominator.

Example: Find $\frac{1}{6}$ of 30.

Step 1. One sixth means 6 equal groups and I take one — so divide 30 by 6.

Step 2. $30 \div 6 = 5$, so $\frac{1}{6}$ of 30 is **5**

Try it: Find $\frac{1}{7}$ of 28.

check: 4

Skill 2 — Any fraction of a number

Plan: to find $\frac{a}{d}$ of a number, first find $\frac{1}{d}$ of it (divide by the denominator), then take a of those groups (multiply by the numerator).

Example: Find $\frac{3}{4}$ of 20.

Step 1. First find one fourth by dividing, then take three of those groups.

Step 2. $20 \div 4 = 5$, and $3 \times 5 = 15$, so $\frac{3}{4}$ of 20 is **15**

Try it: Find $\frac{2}{5}$ of 35.

check: 14

Skill 3 — Finding the whole from a part

Plan: this is the backward question. If you know what a fraction of the number is, first find what *one* group holds, then multiply by the number of groups to rebuild the whole.

Example: $\frac{2}{3}$ of a number is 12. What is the number?

Step 1. The 12 fills 2 of the 3 equal groups, so one group is 12 divided by 2, and the whole is 3 groups.

Step 2. $12 \div 2 = 6$ in each group, and $3 \times 6 = 18$, so the number is **18**

Try it: $\frac{1}{5}$ of a number is 9. What is the number?

check: 45

(!) Watch out: “of” does not always mean multiply the two numbers you see. $\frac{1}{4}$ of 12 is $12 \div 4 = 3$, not 12×4 . Drawing the equal groups keeps the two moves straight.